



COMP10002: Data Warehouse Environment

Investigation of Visual Data Discovery

Paris and Barcelona Airbnb Analysis

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Declaration

This report is submitted in partial fulfilment of the requirements for the module COMP10002: Data Warehouse Environment at the University of the West of Scotland.

I declare that this report embodies the results of my own work and that it has been composed by myself. Following normal academic conventions, I have made do acknowledgement to the work of others.

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(a) Visual Data Discovery Definitions

Definition 1 — Cindi Howson (2013)

Howson (2013, p.57) defines a VDD tool as one that "speeds the time to insight through the use of visualisations, best practices in visual perception, and easy exploration", supporting "business agility and self-service BI through innovations that may include in-memory processing and mashing of multiple data sources." Written for BI professionals, this definition focuses on what VDD delivers for an organisation — speed, self-service, and data integration. The emphasis on in-memory processing signals a practitioner standpoint where performance matters. It is the most technically detailed of the three definitions.

Definition 2 — Tableau (2022)

Tableau (2022) defines data visualisation as "the graphical representation of information and data" providing "an accessible way to see and understand trends, outliers and patterns in data." As a leading VDD vendor, Tableau's definition is deliberately simple and commercial in purpose. There are no mentions of in-memory processing or business agility — the focus is entirely on accessibility. It is aimed at potential customers with little data background, which explains why depth is sacrificed for simplicity.

Definition 3 — IBM (2025)

IBM (2025) defines data discovery as "the process of collecting, evaluating and exploring data from multiple, often disparate, sources", generating insights to "inform and streamline business decisions." Unlike the other two, IBM frames VDD as part of a wider pipeline rather than a standalone tool. This is the broadest of the three definitions and is aimed at enterprise IT professionals, situating visualisation as the output of a larger analytical process.

Comparison and Best Definition

All three definitions agree that VDD makes data more accessible and supports insight discovery. However, they differ in scope and depth. Howson's definition is the most practically grounded and vendor-neutral. Tableau's is the most accessible but also the most superficial. IBM's is the most comprehensive but least focused on the visual exploration experience. Howson's (2013) is the best definition — it captures speed, self-service, and multi-source integration, which are precisely the features that make VDD valuable to organisations.

(b) The Visual Data Discovery Market

Market Size and Growth

The VDD market has grown rapidly in recent years. According to The Business Research Company (2026), the global market reached \$14.51 billion in 2025 and is forecast to grow to \$16.99 billion in 2026 at a CAGR of 17.1%, reaching \$29.11 billion by 2030. North America was the largest region in 2025, whilst Asia-Pacific is the fastest growing. This growth reflects increasing recognition that data-driven decision-making is essential for competitive organisations.

Key Vendors and Market Leaders

The VDD market is dominated by a small number of large technology vendors. According to Solanki (2025), Microsoft Power BI leads with approximately 20% market share, followed by Tableau (Salesforce) at 16.4% and Qlik at 10%. The Gartner Magic Quadrant (2025) places Microsoft, Google (Looker), Oracle, Qlik, Salesforce (Tableau) and ThoughtSpot in the Leaders quadrant. Gartner (2025) also notes that modern ABI platforms are "increasingly leveraging generative AI to enhance productivity", fundamentally changing how VDD tools are used.

Factors Driving Growth

Several factors are driving VDD market growth. The Business Research Company (2026) identifies increasing enterprise data volumes, rising BI tool adoption, and growing demand for faster decision making as the primary historic drivers. The democratisation of analytics — making data insights accessible to all employees rather than just specialists — is also driving demand for self-service tools. As Tableau (2024) describes, visual analytics allows users to "ask a question, get the answer, and ask follow-up questions — all within a visual interface." Looking ahead, cloud-native platforms and AI-assisted exploration are forecast to be the dominant trends through 2030.

(c) The Visual Data Discovery Process

(i) An Investigation of Open Data

Open data refers to data made freely available for anyone to access, use, and share. The European Commission's Open Data Portal (2024) defines it as "information collected, produced or paid for by the public bodies and made freely available for re-use for any purpose." Open data is produced by governments, international organisations, NGOs, academic institutions, and independent community projects. The movement has grown due to technological advances that made publishing large datasets cheap and accessible, increasing public demand for government transparency, and the recognition that shared data drives economic innovation. The European Commission (2024) estimates the EU open data market grew from 55.3 billion EUR in 2016 to 75.7 billion EUR by 2020, creating nearly 100,000 jobs.

(ii) Visual Data Discovery Using Open Data — Paris and Barcelona

City Selection

Paris and Barcelona were selected for personal and analytical reasons. I have visited Paris before and was curious about its accommodation landscape. My sister visited Barcelona and described it as an incredibly vibrant city, which made me want to explore its Airbnb activity. Analytically, both are major European tourist destinations with large active Airbnb markets and an interesting regulatory contrast — Paris enforces strict 120-night annual limits on short-term rentals, whilst Barcelona has developed its own regulatory approach. These differences are clearly reflected in the data.

Data Source and Preparation

Data was sourced from Inside Airbnb (2024), a mission-driven project that scrapes and publishes Airbnb data to support housing advocacy. The Paris dataset contained 82,216 listings across 18 columns; Barcelona contained 18,198 listings across 19 columns. One significant limitation was that the price column was empty in both datasets — a known issue with recent Inside Airbnb data. Analysis therefore focused on availability, minimum nights, number of reviews, and host listings count. Both datasets were examined in Excel first, then imported into Tableau Public.

Statistic	Minimum Nights	Number of reviews	Availability 365	Calculated Host Listings Count
Mean	16.59250701	11.20063817	209.5721516	68.49023491
Median	3	1	245	9
Mode	1	0	0	1
Std Deviation	31.5898986	24.99937419	130.3136958	129.1951186
Variance	997.9216934	624.96871	16981.65931	16691.37866
Min	1	0	0	1
Max	1124	938	365	549
Range	1123	938	365	548
Skewness	14.32611619	10.90138158	-0.434266223	2.36157057
Kurtosis	339.2110814	277.0069595	-1.28648643	4.875949262

Figure 1: Descriptive statistics for Barcelona — Microsoft Excel

Statistic	Paris Min Nights	BCN Min Nights	Paris Availability	BCN Availability
Mean	44.04735318	27.22796965	127.2531001	23.15124675
Median	3	7	74	1
Mode	2	0	0	1
Std Deviation	110.0782549	61.90759889	135.2151008	92.45202403
Variance	12117.22221	3832.5508	18283.12349	8547.376748
Min	1	0	0	1
Max	1000	4040	365	816
Range	999	4040	365	66.92
Skewness	2.572065327	10.43833971	0.499107797	6.849191993
Kurtosis	4.914748762	341.3941732	-1.374904415	52.63695843

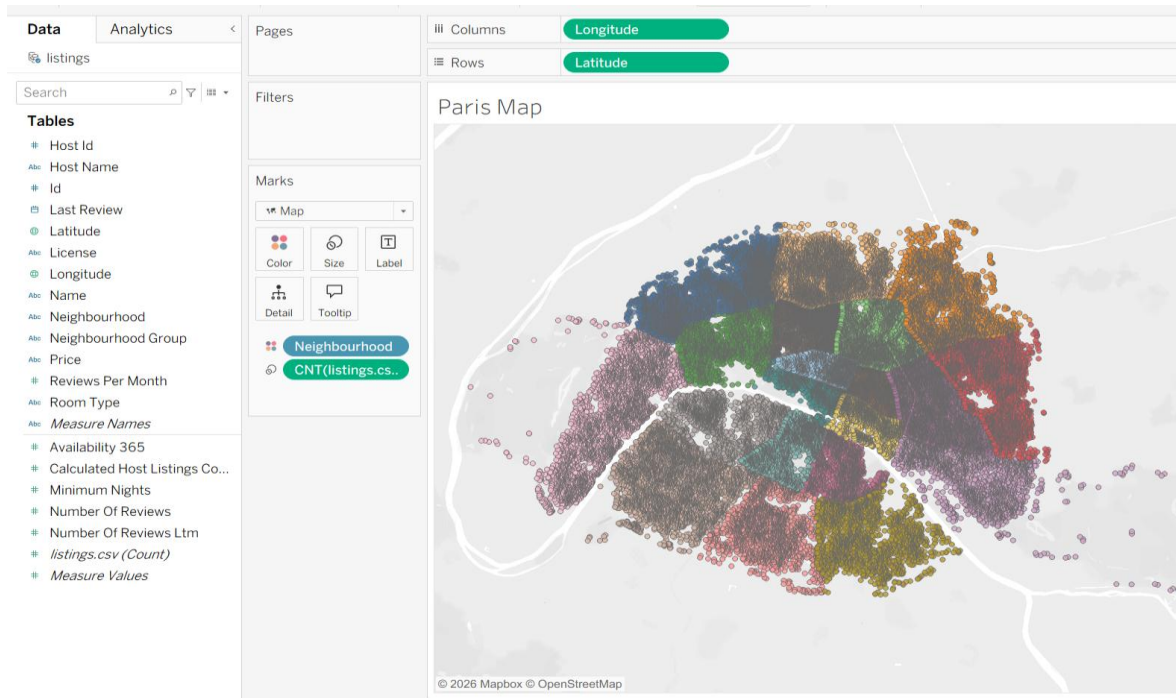
Figure 2: Descriptive statistics for Paris — Microsoft Excel

The statistics reveal several interesting patterns. Paris has a mean minimum nights of 44.05 compared to Barcelona's 16.59, yet both share a median of 3 — the large mean/median gap in Paris reflects a small number of listings with very high minimum night requirements, likely driven by French regulations. For availability, Barcelona's mean (209.57 days) is considerably higher than Paris (127.25 days), suggesting a more commercially oriented hosting culture in Barcelona. The negative skewness for Barcelona availability (-0.43) further supports this, indicating the distribution skews towards higher availability.

Visualisations

Visualisation 1 — Geographic Distribution of Paris Listings

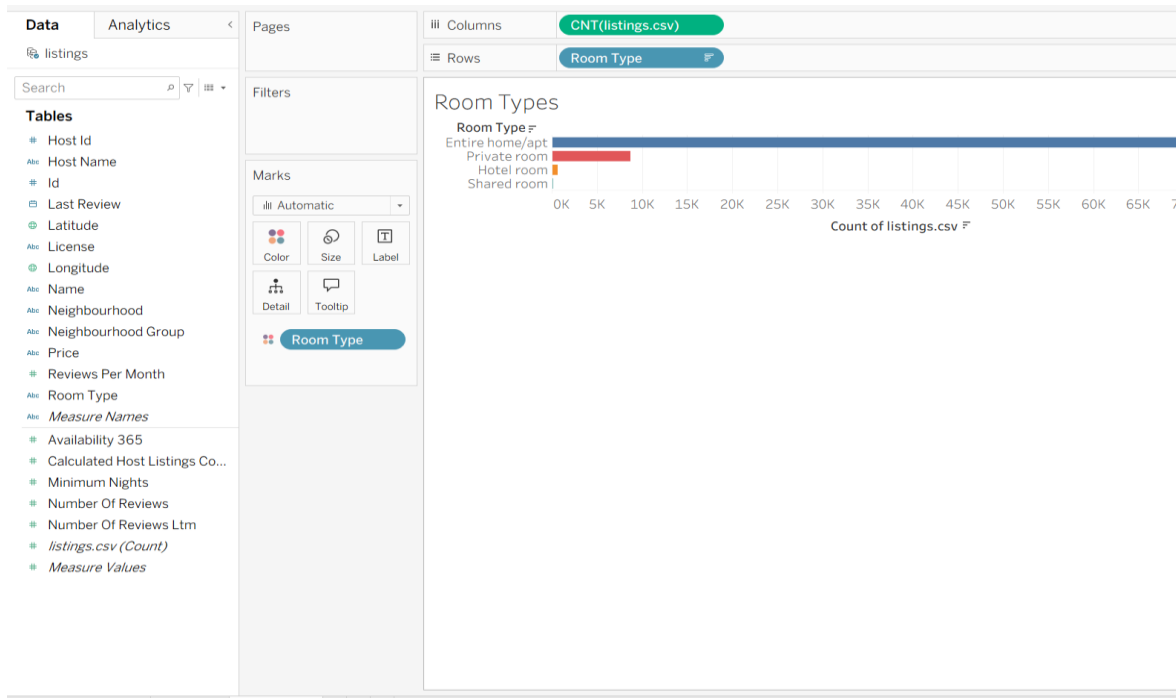
The geographic map plots every Paris listing by latitude and longitude, coloured by neighbourhood. The central and northern arrondissements show the highest listing density, consistent with their proximity to major tourist attractions. This is directly useful for local government officers assessing which neighbourhoods face the greatest housing pressure from short-term rentals.



Visualisation 1: Geographic map of Paris Airbnb listings coloured by neighbourhood

Visualisation 2 — Room Type Breakdown (Paris)

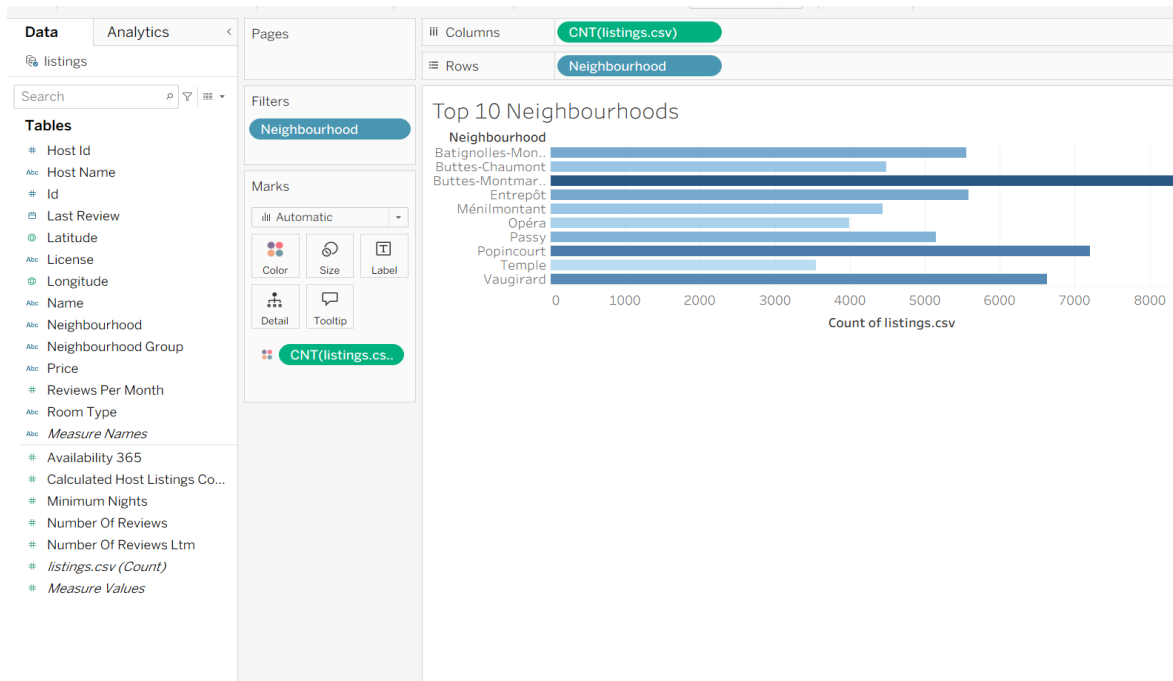
Entire home listings account for approximately 72,297 of Paris's 81,853 total listings (88%). Private rooms account for 8,727, whilst hotel and shared rooms are negligible. This suggests a substantial proportion of Paris properties have been removed from the residential housing market to serve tourists full-time.



Visualisation 2: Room type breakdown for Paris listings

Visualisation 3 — Top 10 Paris Neighbourhoods

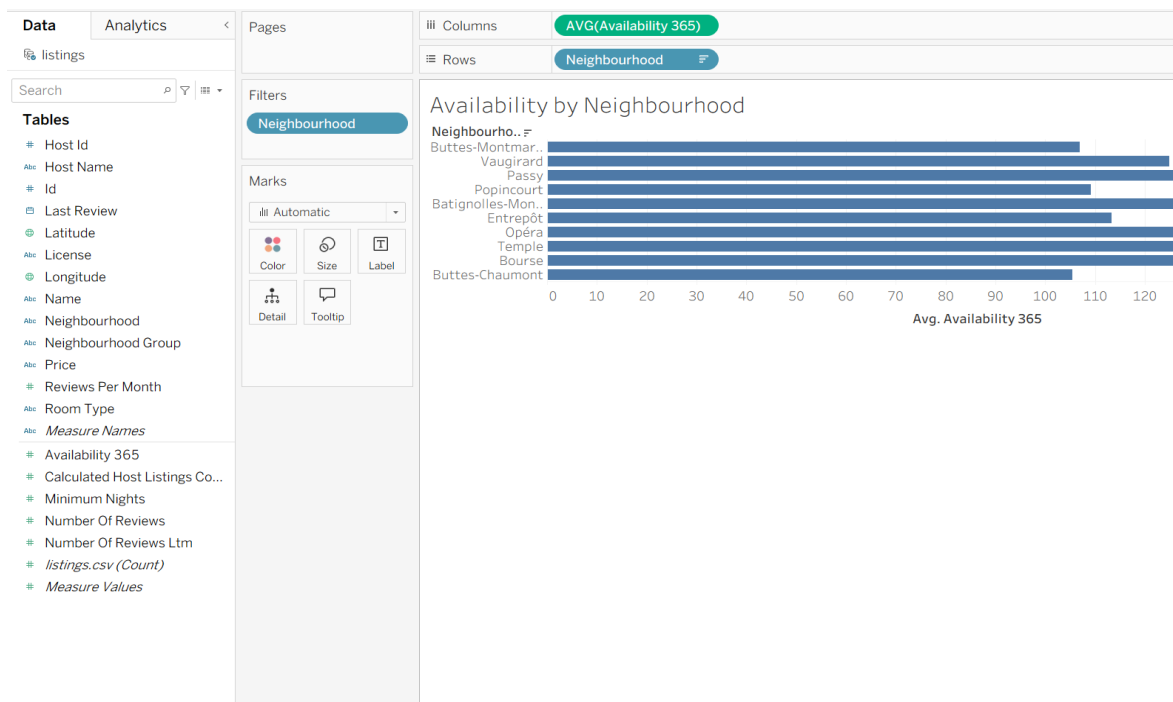
Buttes-Montmartre has the highest number of listings (approximately 8,500), followed by Popincourt and Vaugirard. These are popular tourist areas in northern and central Paris. High listing concentration in these areas has clear implications for housing affordability for local residents.



Visualisation 3: Top 10 Paris neighbourhoods by number of listings

Visualisation 4 — Average Availability by Neighbourhood

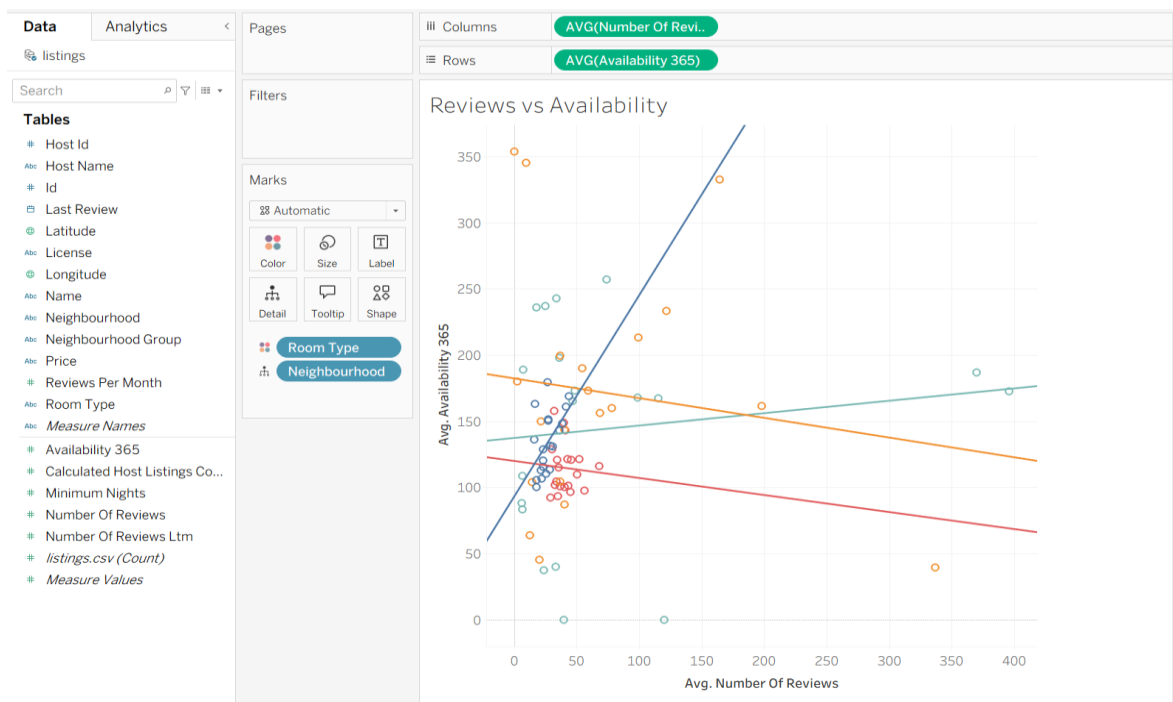
The top Paris neighbourhoods all show high average availability, with most exceeding 600 total days across listings. This suggests listings are actively managed for large portions of the year, consistent with commercial short-term rental operations rather than occasional hosting.



Visualisation 4: Average availability by neighbourhood — top 10 Paris neighbourhoods

Visualisation 5 — Reviews vs Availability

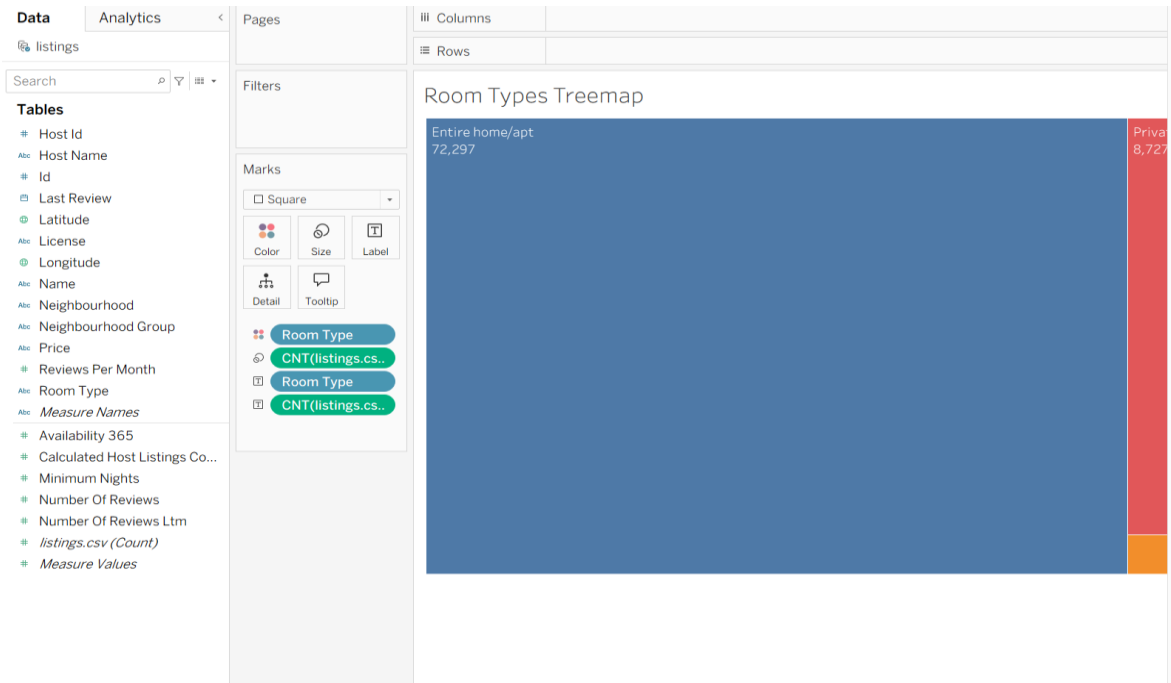
The scatter plot shows a slight negative relationship — listings with more reviews tend to have slightly lower availability, suggesting highly reviewed listings are booked more frequently. Most listings cluster in the low-reviews, moderate-availability area, with a small number of outliers showing very high review counts.



Visualisation 5: Scatter plot — average reviews vs average availability coloured by room type

Visualisation 6 — Room Types Treemap (Paris)

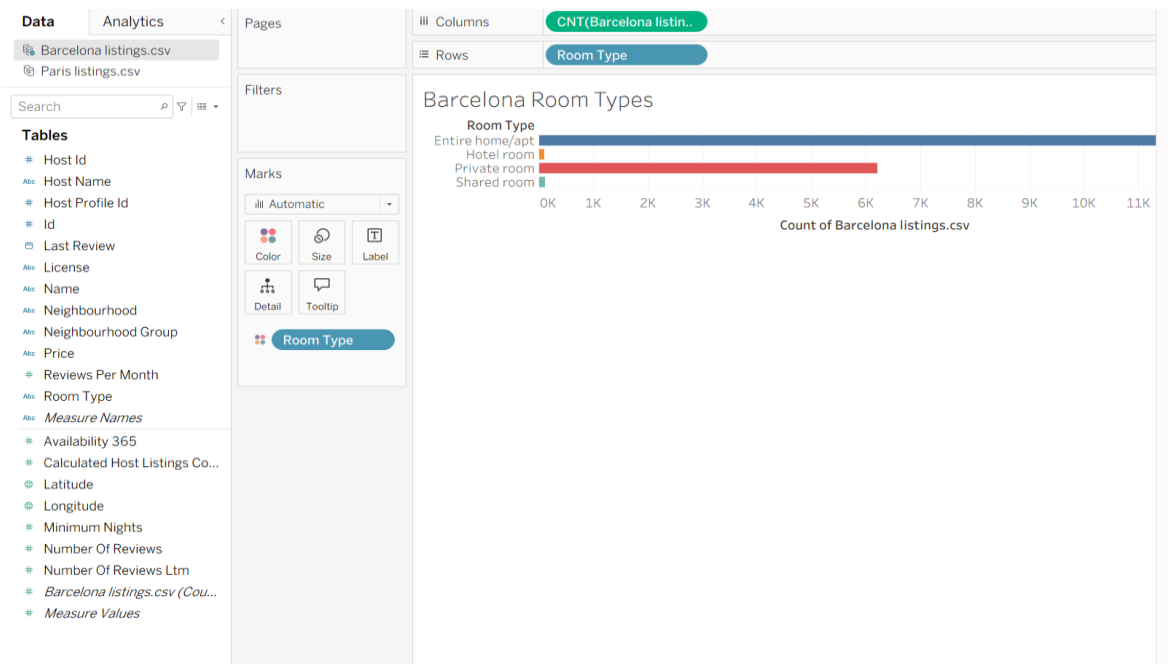
The treemap makes the dominance of entire home listings even more visually immediate than the bar chart — the entire home square occupies nearly all available space. This type of proportional visualisation communicates scale effectively to audiences unfamiliar with bar charts or tables.



Visualisation 6: Treemap showing proportional breakdown of room types in Paris

Visualisation 7 — Barcelona Room Types

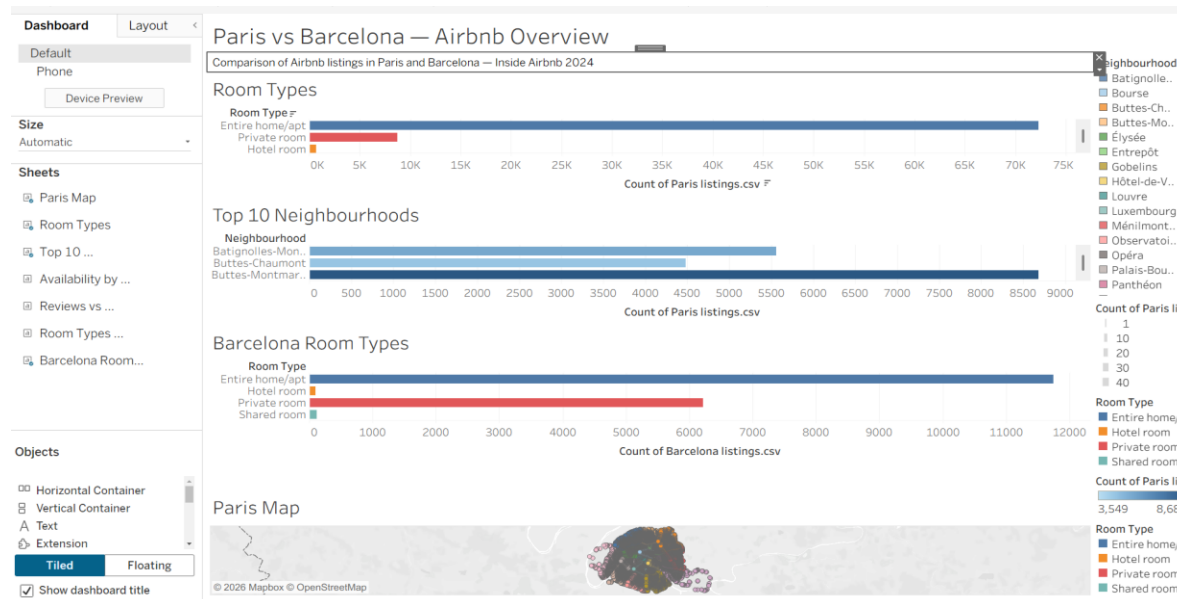
Barcelona shows a similar pattern to Paris with entire homes dominating (approximately 12,000 of 18,198 listings), but with a notably stronger proportion of private rooms and a more visible hotel room category. Combined with the higher mean availability (209 days vs Paris's 127), Barcelona's market appears even more commercially intensive on a per-listing basis.



Visualisation 7: Room type breakdown for Barcelona listings

Visualisation 8 — Dashboard

The dashboard combines the Paris map, room type comparisons for both cities, and the top 10 neighbourhoods chart into a single overview. The side-by-side room type charts highlight differences between the two cities effectively, whilst the map provides geographic context. For Airbnb management or local government purposes, this provides a practical starting point for understanding both markets at a glance.



Visualisation 8: Dashboard — Paris vs Barcelona Airbnb Overview

(d) Evaluating Tableau

System Usability Scale Assessment

To evaluate Tableau's usability, the System Usability Scale (SUS) developed by Brooke (1996) was applied after completing the Paris and Barcelona analysis tasks. SUS is a ten-item Likert scale producing a composite usability score between 0 and 100. Brooke (1996, p.189) describes it as a "reliable, low-cost usability scale that can be used for global assessments of systems usability." It was selected because it is quick, vendor-neutral, and widely validated. The total score contribution from all ten items was 24, giving a final SUS score of $24 \times 2.5 = 60$. This sits just below the commonly cited threshold of 68, which is generally considered the boundary between acceptable and good usability.

Interpretation and Opinion

Tableau is a powerful and visually impressive VDD tool, positioned as a Leader in the Gartner Magic Quadrant (2025). However, the learning curve for a new user is genuinely steep. Creating the geographic map required understanding the difference between dimensions and measures, and converting latitude and longitude from averaged values to individual dimensions — a step that is not obvious without guidance. The process of building charts across separate worksheets before assembling a dashboard is also more complex than competing tools such as Power BI. Solanki (2025) notes that organisations should expect three to six months for full team proficiency, which is consistent with the experience of building these visualisations. That said, the quality of outputs — particularly the geographic map and dashboard — is impressive once the workflow is understood. A SUS score of 60 is an honest reflection for a first-time user.

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